

RESEARCH ARTICLE

Preparation of porous poly(4-tert-butylstyrene) based monoliths with high efficiency for oil-water separation via high internal phase emulsion template

Yulai Zhao^{1,2,3}  | Lianzhi Shen¹ | Yu Yuan¹ | Longqiang Xiao^{1,2,3} | Jingyu Cai^{1,2,3} | Zhen Lu² | Linxi Hou^{1,2,3}

¹Department of Materials-Oriented Chemical Engineering, College of Chemical Engineering, Fuzhou University, Fuzhou, China

²Fuzhou University, Qingyuan Innovation Laboratory, Quanzhou, China

³Fujian Key Laboratory of Advanced Manufacturing Technology of Specialty Chemicals, Fuzhou, China

Correspondence

Yulai Zhao and Linxi Hou, Department of Materials-Oriented Chemical Engineering, College of Chemical Engineering, Fuzhou University, Fuzhou 350116, China.
Email: yulai_zhao@fzu.edu.cn and lxhou@fzu.edu.cn

Funding information

Natural Science Foundation of Fujian Province, Grant/Award Number: 2022J01569; National Natural Science Foundation of China, Grant/Award Number: U21A200353; Major Program of Qingyuan Innovation Laboratory, Grant/Award Number: 00122004; Key Program of Qingyuan Innovation Laboratory, Grant/Award Number: 00221003; '111' Program of Fuzhou University

Abstract

Developing low-cost and highly efficient hydrophobic absorbents via simple and environmentally friendly method is demanded for oil-water separation. This paper reports the preparation of hydrophobic absorbents of porous poly(4-tert-butylstyrene-co-divinylbenzene) (poly(*t*BS-co-DVB)) via high internal phase emulsion (HIPE) template method. The effects of crosslinking degree and internal phase volume fraction on the porous structure and adsorption performance of the absorbents are investigated. As the *t*BS/DVB molar ratio increases from 1:1 to 4:1, the adsorption performance of poly(*t*BS-co-DVB) toward hexane first increases to a maximum at a molar ratio of 3:1 and then decreases. As the internal phase volume fraction increases, the specific surface area (SSA) and porosity of the poly(*t*BS-co-DVB)s notably increase from 33.06 m²/g and 84.07% to 102.89 m²/g and 92.94%, respectively. When the internal phase fraction is 95 vol%, the obtained poly(*t*BS-co-DVB) foam with a water contact angle of 140° shows an absorbing capacity as high as 70.21 g/g for dichloromethane. After 10 consecutive adsorption-desorption cycles, the adsorption performance of the poly(*t*BS-co-DVB) foam shows only a slight decrease, indicating an excellent recycling ability. The obtained poly(*t*BS-co-DVB) foams exhibit the advantages of facile preparation, high oil absorption capacity and satisfactory recoverability, which enable them potential as sustainable oil adsorbents.

KEYWORDS

HIPE, oil adsorption, poly(*t*BS-co-DVB) foam, porous monolith

1 | INTRODUCTION

With the development of the economy, modern society is increasingly dependent on oil products.^{1–3} However, oil pollutions happen constantly because of oil leakages during transportation and industrial production. Every year, a large amount of oil flows into the oceans, which may

cause a serious impact on the ecological environment if not treated adequately in time.^{4–6} Also, the oil extraction process generates a large amount of oily wastewater, which needs to be treated to avoid environmental pollution before discharge. To solve these problems, many methods have been proposed including dispersion,⁷ absorption,⁸ filtration,⁹ bioremediation¹⁰ and burning.¹¹

Among these methods, absorption with highly porous materials is considered to be one of the most promising methods owing to the advantages of convenience, inexpensive, high efficiency and no secondary pollution.^{12–14}

In a range of oil absorbing materials, three-dimensional (3D) monolithic porous materials are considered as expectative materials for oil absorbing due to their large specific surface area, low density, high porosity and easy recovery.^{15–17} Commonly used 3D monolithic porous materials mainly include aerogels,^{18–20} sponges,^{21–23} and foams.^{24–26} These materials have interconnected pore structures that can absorb oil via capillary action, and can swell as they absorb oil, allowing the internal voids to absorb more oil. In contrast, granular and powdered materials tend to agglomerate during actual adsorption, resulting in a lower adsorption surface area and thus a lower adsorption capacity and efficiency.²⁷ In addition, powder materials are difficult to collect and recover after adsorption, which limits their application to actual oil contamination.²⁸

However, the original sponge is amphiphilic and absorbs water and oil simultaneously, which would reduce the adsorption capacity and fail to achieve the purpose of efficient separation of water and oil.²⁹ In addition, conventional aerogels such as silica aerogels are not ideal for oil removal due to their inherent brittleness.³⁰ As a result, complex surface modifications and processing are required to overcome these difficulties. For instance, Xu et al.³¹ modified the melamine sponge with cobalt oxyhydroxide by hydrothermal method and then with stearic acid. The surface modified sponge changed from hydrophilic to superhydrophobic, with a water contact angle of 155°. He et al.³² prepared a superplastic and superhydrophobic bacterial cellulose/silica aerogel by vacuum infiltration and freeze-drying using a 3D self-assembled cellulose backbone to reinforce phase and methyltriethoxysilane derivatives as fillers. The aerogel can withstand up to 80% compressive strain with a water contact angle of 152°. It can be concluded that the modification processes mentioned above are complex and energy-intensive, which made it hard to apply to practical applications.

HIPEs refer to highly viscous emulsions with an internal phase volume fraction of more than 74%.^{33,34} The obtained porous polymer (polyHIPE) foams with interconnected porous structures are obtained by polymerizing the continuous phase and removing the dispersed phase.³⁵ Based on this property, hydrophobic and lipophilic polymer foams can be prepared by choosing hydrophobic monomer as the continuous phase for polymerization and aqueous phase as the dispersed phase. From this a conclusion can be drawn that the choice of functional monomers is crucial, as it determines

the chemical composition of the material surface which in turn affects the hydrophobicity of the material. Wang et al.³⁶ prepared hydrophobic fluoro-polyHIPEs with water contact angles up to 148.6° by using fluorinated monomer 2,2,2-trifluoroethyl methacrylate as hydrophobic monomer and polylactic acid as co-stabilizer. The prepared porous material has an excellent oleophilicity. However, emulsion containing fluoro-monomer usually require the use of special surfactants to stabilize, which increases the complexity of the experiment. Therefore, in this paper the styrene-derived monomer 4-tert-butylstyrene (*t*BS) was used as hydrophobic monomer. The poly(*t*BS-*co*-DVB) foam will contain more alkyl chains, which will further enhance the hydrophobicity of the prepared polyHIPEs. In addition, the interconnected pore structure is very important for oil adsorption materials. Therefore, the porogenic agent of molecular toluene was introduced into the HIPE system to build more developed porous structures. Molecular toluene can be removed from the poly(*t*BS-*co*-DVB) backbone by soxhlet extraction to form abundant interconnected pore structure.

Herein, HIPE template method was applied to fabricate hydrophobic and lipophilic 3D foams in one step for efficient oil absorbing and oil-water separation, without any surface modification. In this method, 4-tert-butylstyrene (*t*BS) was added to the continuous phase containing divinylbenzene (DVB) and toluene. Then, poly(*t*BS-*co*-DVB) foams with excellent oil adsorption capacity and recyclability were prepared by polymerizing the continuous phase followed with internal phase removal.

2 | EXPERIMENTAL

2.1 | Materials

4-tert-butylstyrene (*t*BS), divinylbenzene (DVB, 55% mixture of isomers), sorbitan monooleate (Span 80), Sudan II, sodium hydroxide (NaOH) and anhydrous calcium chloride (CaCl₂) were provided by Shanghai Aladdin Biochemical Technology Co., Ltd. 2, 2-Azobis(isobutyronitrile) (AIBN) was purchased from TCI (Shanghai) Development Co., Ltd. Toluene was provided by Sino-pharm Chemical Reagent Co., Ltd. Chloroform, n-hexane, ethyl acetate and petroleum ether were provided by Tianjin Zhiyuan Chemical Reagent Co., Ltd. Peanut oil was provided by Laiyang Luhua Intense Peanut Oil Co. Engine oil was provided by Royal Dutch Shell. Deionized water was used in the work thoroughly. DVB and *t*BS were firstly washed three times with 10 wt% aqueous NaOH to remove the inhibitor and then washed with

TABLE 1 Parameters of the PolyHIPEs.

Sample ^a	<i>t</i> BS (g)	DVB (g)	W/(W + O) ^b (V/V)	<i>t</i> BS:DVB (molar ratio)
polyHIPE-C1	0.2760	0.2240	80%	1:1
polyHIPE-C2	0.3556	0.1444	80%	2:1
polyHIPE-C3 ^c	0.3934	0.1066	80%	3:1
polyHIPE-C4	0.4156	0.0844	80%	4:1
polyHIPE-80% ^c	0.3934	0.1066	80%	3:1
polyHIPE-85%	0.3934	0.1066	85%	3:1
polyHIPE-90%	0.3934	0.1066	90%	3:1
polyHIPE-95%	0.3934	0.1066	95%	3:1

^aFor all the samples, the dosage of surfactant Span 80 is 0.2 g. Toluene was used as porogenic agent (0.5 mL).

^bVolume ratio of the CaCl₂ aqueous solution and CaCl₂ aqueous solution plus oil phase (*t*BS, DVB and toluene).

^cThe samples are prepared in the same way but named differently for comparison: polyHIPE-C3 = polyHIPE-80%.

deionized water to remove the NaOH and dried with anhydrous calcium chloride. Others were used directly without further purification.

2.2 | Preparation of polyHIPEs

During the preparation process, the effects of the molar ratio of monomer to crosslinker and the volume fraction of the internal phase on the pore structure of porous polymer foam were investigated. Under the conditions of the total mass of *t*BS and DVB is 0.5 g and the volume fraction of internal phase is 80%, the polymer foams prepared with molar ratio of *t*BS to DVB = 1:1, 2:1, 3:1, 4:1 are named as polyHIPE-C1, polyHIPE-C2, polyHIPE-C3, polyHIPE-C4, respectively. After determining the optimal molar ratio of *t*BS to DVB under the internal phase volume fraction of 80%, the polymer foams with internal phase volume fractions of 85%, 90%, and 95% were then prepared in order to further investigate the effect of internal phase volume fraction and named as polyHIPE-85%, polyHIPE-90%, polyHIPE-95% respectively. More information about polymer foam composition can be found in Table 1. The detailed experimental steps are as follows. First, 0.015 g of initiator AIBN and 0.2 g of surfactant Span 80 were added to the oil phase containing of 0.5 g organic phase and 0.5 mL porogenic agent of toluene in the centrifugal tube. After total dissolution, the above solution was transferred to a 25 mL two-necked flask equipped with a stirrer. Calcium chloride aqueous solution as the dispersed phase was slowly added into the two-necked flask at a stirring speed of 1100 r/min. After completing the drop of dispersed phase, the stirring is maintained for another 1.5 h to form a viscous emulsion which was then sealed in a centrifuge tube and polymerized in an oven at 70°C for

24 h. Then residual surfactants and unreacted monomers were removed via soxhlet extraction using absolute ethanol at 90°C for 12 h. Finally, the extracted polymer foam was put into a vacuum oven at 60°C for 12 h to obtain hydrophobic porous foams.

2.3 | Characterization

Morphology and microstructure of hydrophobic porous polymer foams were examined by scanning electron microscopy (SEM, S-4800, Hitachi, Japan). Infrared spectrometry (FTIR, Nicolet iS50, Thermo Fisher Scientific, China) was used to examine the chemical bonds and functional groups of polymer foams. The water contact angle (WCA) of porous polymer foam was measured by contact angle instrument (CA, HARKE-SPCAX1, Beijing Hake Test Instrument Factory, China). Mercury press testing adopted by mercury porosimetry (MIP, AutoPore9510, American) is used to characterize the density, porosity and pore size distribution of polymer foam. The materials' thermal stability was tested by thermal gravimetric analysis (TGA, */STA449C/6/G, German).

2.4 | Oil adsorption performance test

A sample with a mass of about 0.1 g was cut from the polymer foam and weighed, and its weight was recorded as m_0 . Then put it into a beaker with 15 mL of oil or organic solvent, and when the adsorption saturation was reached, the polymer foam was removed from the beaker and its mass was weighed and recorded as m_1 . The oil adsorption performance of the material was calculated by the following equation:

$$\text{Absorption capacity} = \frac{m_1 - m_0}{m_0} \text{ g/g}$$

Three parallel experiments were conducted for each test, and the final adsorption capacity was the average of the three experiments. In order to test the oil-water separation properties of the prepared polymer foam, ethyl acetate and dichloromethane, which has a lower density and a higher density than water, were stained with Sudan II in order to facilitate observation. Then stained oil was dropped in a beaker containing water for the actual oil-water separation process of the samples. To test the oil adsorption cycle stability, the adsorbed saturated sample was regenerated by centrifugation at 3000 r/min for 4 min. Then its weight was weighed and recorded as m_2 , then the porous polymer foam was subjected to a continuous adsorption-desorption cycle test. The oil retention rate of the material was obtained by the following equation:

$$\text{Oil retention} = \frac{m_1 - m_2}{m_1} \text{ g/g}$$

The centrifuged samples were used directly for oil adsorption cycle stability testing.

3 | RESULTS AND DISCUSSION

The process of preparing hydrophobic porous polymer foams by high internal phase emulsion (HIPE) template method is shown in Scheme 1. Firstly, a certain amount of CaCl_2 aqueous solution was added dropwise to the oil phase containing *t*BS, crosslinker and porogenic agent under high-speed stirring conditions. With the addition

of calcium chloride aqueous solution, the emulsion system became more and more viscous. The viscous high internal phase emulsion was vigorously stirred at 1100 rpm for 90 min to form a homogeneous emulsion. The organic phase is polymerized with the subsequent removal of the aqueous phase to form porous structures, by which porous polymer foams are obtained and named as polyHIPEs.

3.1 | Chemical structure of polyHIPEs

The chemical composition of prepared polyHIPEs, crosslinker (DVB) and hydrophobic monomer (*t*BS) were

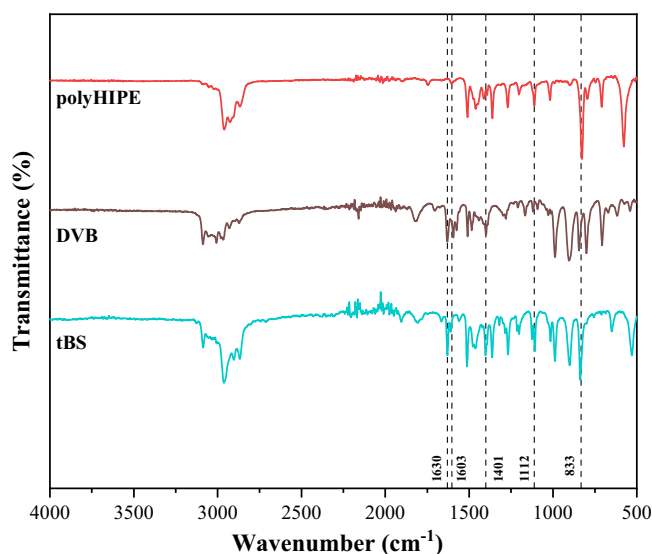
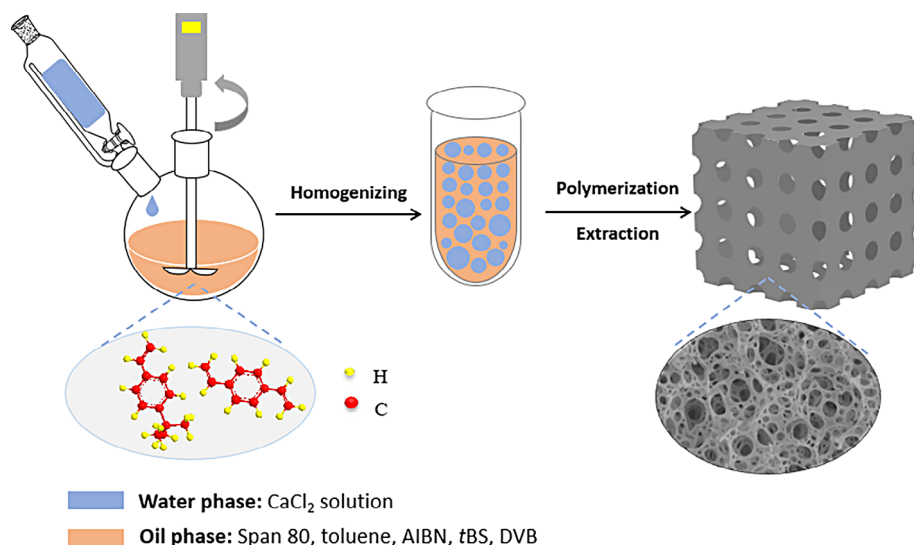


FIGURE 1 FTIR spectra of polyHIPE, DVB, *t*BS. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)] [https://onlinelibrary.wiley.com/terms-and-conditions]



SCHEME 1 Schematic illustration of the preparation of hydrophobic polyHIPE via HIPE template [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)] [https://onlinelibrary.wiley.com/terms-and-conditions]

examined using FTIR spectroscopy. As shown in Figure 1, the absorption peaks at 1603 and 1401 cm^{-1} are the characteristic vibrational absorption peaks of the benzene ring skeleton, and the peak at 1112 cm^{-1} is the out-of-plane vibrational absorption peak of C—H on the benzene ring. The peak at 833 cm^{-1} represents the absorption peak of tert-butyl group of *t*BS. The absence of the peak at 1630 cm^{-1} in the polyHIPE proves that the C=C groups in *t*BS and DVB are fully converted to saturated polymer chains in the polyHIPE, suggesting the successful preparation of the polymer foam. Thermal stability is important for oil adsorption materials. As

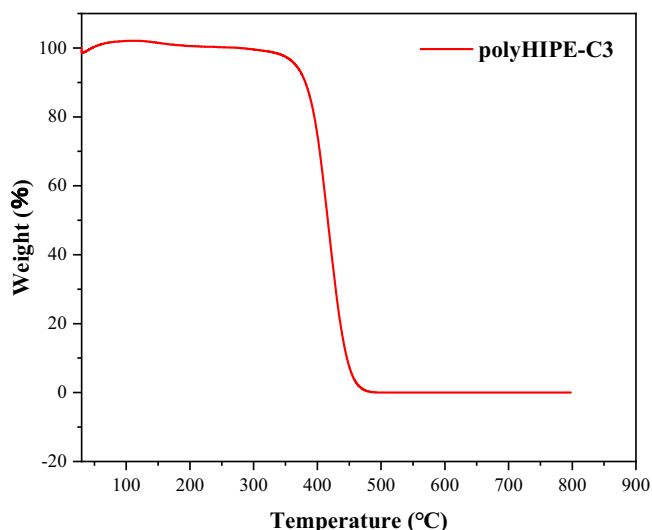


FIGURE 2 TG analysis of polyHIPE-C3 foam. [Color figure can be viewed at wileyonlinelibrary.com]

shown in Figure 2, before the temperature reached 320°C, no significant change in the mass of the prepared polyHIPE was observed. While after the temperature exceeded 320°C, the mass of the polymer foam decreased sharply. These results showed that the polymer foam has excellent thermal stability and can be used for oil-water separation at high temperature.

3.2 | Morphology observation of polyHIPEs

All prepared samples have interconnected porous structures as observed by SEM. It can be seen from Figure 3 that the pore size of the polyHIPEs gradually decrease as the content of DVB increases, which is due to the increase in cross-linkage making the cross-linked network of the emulsion system more compact. The adsorption capacity of hexane was tested as shown in Figure S1, and the adsorption performance showed a trend of increasing and then decreasing with the increasing of *t*BS/DVB ratio and reached the maximum at 3:1, which may be because when the cross-linkage is too high, the pore size of the polyHIPE is small, resulting in low specific surface area. When the ratio reaches to 4:1, the structure of the polyHIPE with low cross-linkage is partially destroyed due to its easy dissolution in organic solvents. Therefore, the optimal ratio is 3:1. In addition, the effect of the internal phase volume fraction is investigated in a similar way. From Figure 4, it is notable that the number of void number and the pore size of the polymer foams gradually increase as the volume fraction of

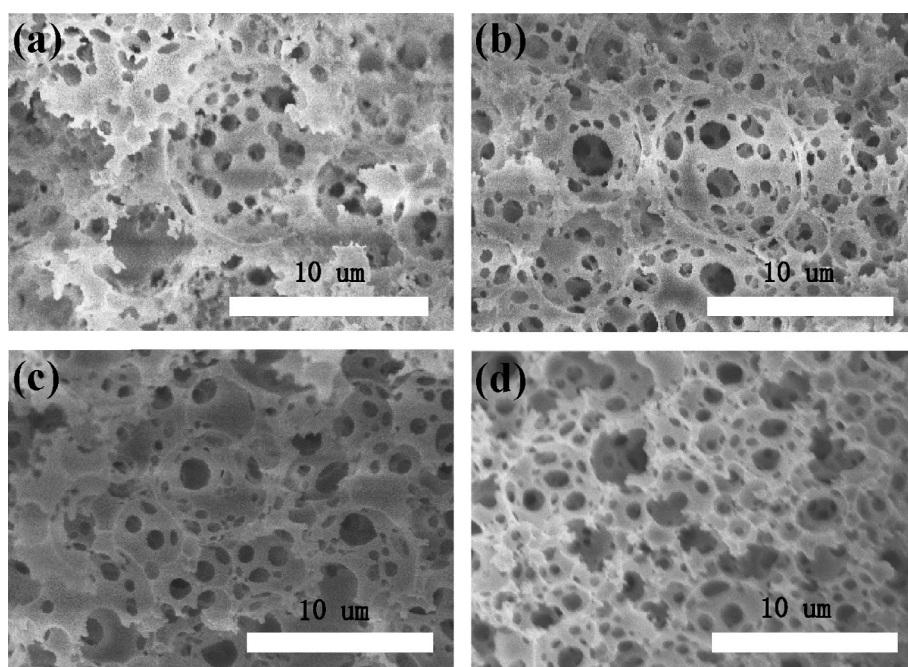


FIGURE 3 SEM images of polyHIPEs with different mole ratios of *t*BS to DVB (a) 1:1 (b) 2:1 (c) 3:1 (d) 4:1.

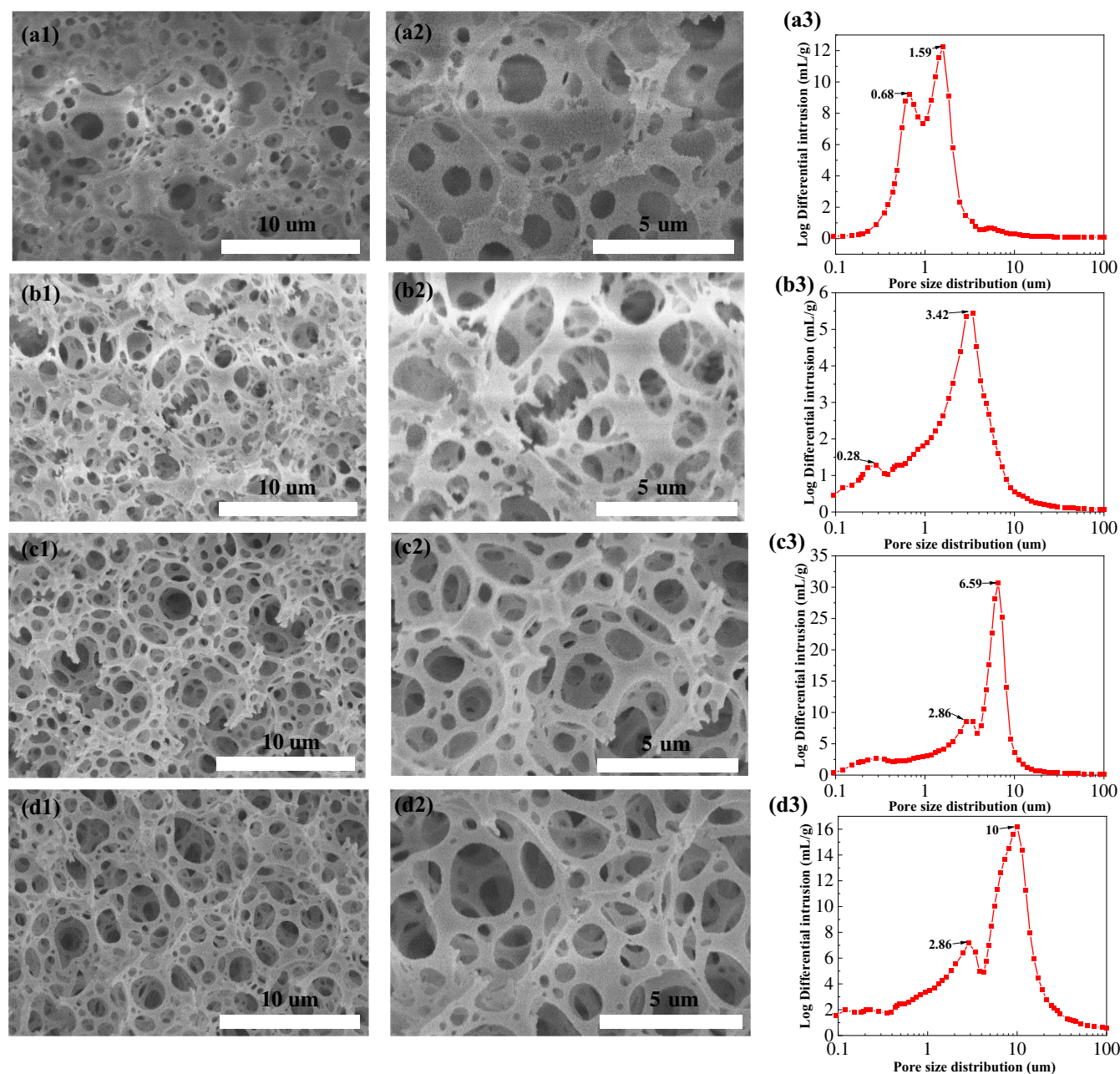


FIGURE 4 SEM images of polyHIPEs obtained with $tBS/DVB = 3:1$ at different internal phase volume fractions (a_1, a_2) 80%, (b_1, b_2) 85%, (c_1, c_2) 90%, (d_1, d_2) 95%; (a_3), (b_3), (c_3) and (d_3) are the pore size distributions of the corresponding polyHIPEs. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 2 Mercury intrusion porosimetry data of PolyHIPEs.

Sample	Foam density (g/cm ³)	Skeletal density (g/cm ³)	Porosity (%)	Specific surface area (m ² /g)
polyHIPE-80%	0.1880	1.1803	84.07	33.06
polyHIPE-85%	0.1197	1.0026	88.06	58.04
polyHIPE-90%	0.0705	0.9935	90.03	63.18
polyHIPE-95%	0.0663	0.9390	92.94	102.89

FIGURE 5 Photographs for (a) water curled into spheres on the surface and (b) Sudan II dyed ethyl acetate adsorbed into the polyHIPE; (c) Water contact angle of polymer foam. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.53801)]

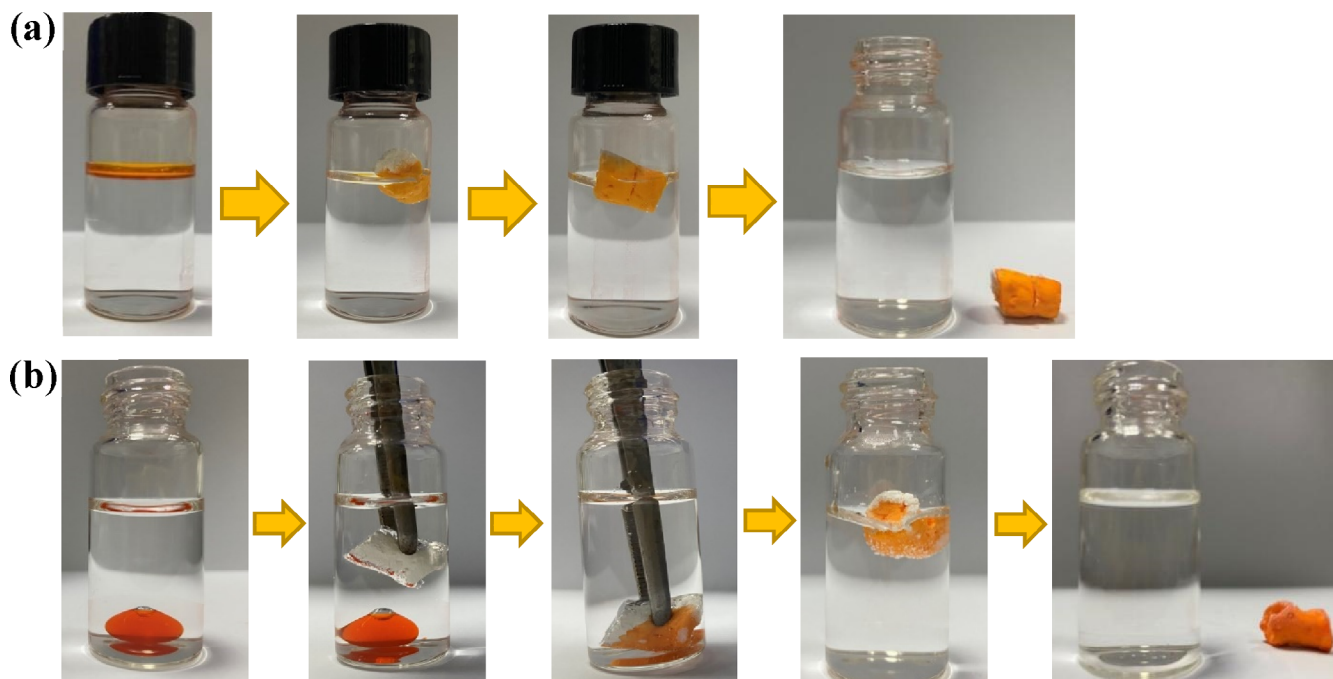
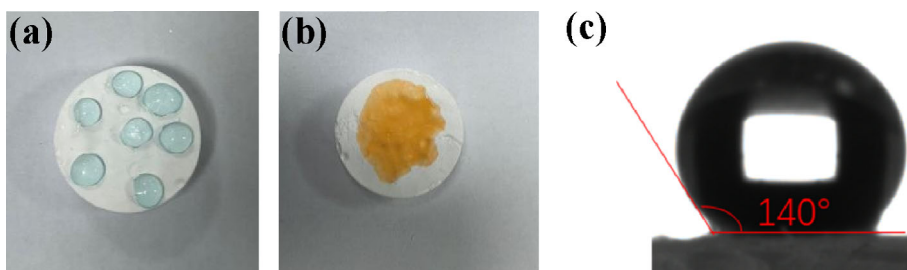


FIGURE 6 Actual oil-water separation process: (a) the removal of ethyl acetate from the water surface by the polyHIPE-95% with 3–5 s, (b) the removal procedures of submerged CH_2Cl_2 . [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.53801)]

dispersed phase increases. This is because as the volume fraction of dispersed phase increases, the concentration of emulsifier in the HIPE system become lower, resulting in a weaker emulsification ability, which makes the polymer foam pore size larger.^{37,38} The increase in the volume fraction also leads to a tighter accumulation of droplets in the emulsion system, and the thickness of the interfacial film around the droplets becomes thinner, leading to the formation of small pores and easy formation of channels between droplets, resulting in an increase in small pore number.

In addition, as the water phase in the emulsion system increased, the density of the polyHIPEs decreased while the porosity and SSA increased. From Table 2 it can be observed that the density and skeletal density of the obtained polyHIPE foams significantly decreased from 0.1880 to 0.0663 g/cm³, 1.1803 to 0.939 g/cm³, respectively and the specific surface area and porosity increased from 33.06 m²/g to 102.89 m²/g, 84.07% to

92.94%, respectively with the corresponding increase of water phase from 80% to 95%. These indicated that the density and porosity of the polymer foam can be regulated by adjusting the content of the water phase. Meanwhile, we can see from the pore size distribution diagram that the pore size distribution of the polymer foam mainly concentrated in 0.68 and 1.59 μm when the content of the aqueous phase is 80%, while the pore size distribution of the polymer foam increases as the content of the aqueous phase continues to increase. Finally, it is concentrated at 2.86 and 10 μm with the aqueous phase of 95%.

3.3 | Wettability and oil-water separation

A series of experiments were designed to study the hydrophobicity of polymer foam. Figure 5a,b show water

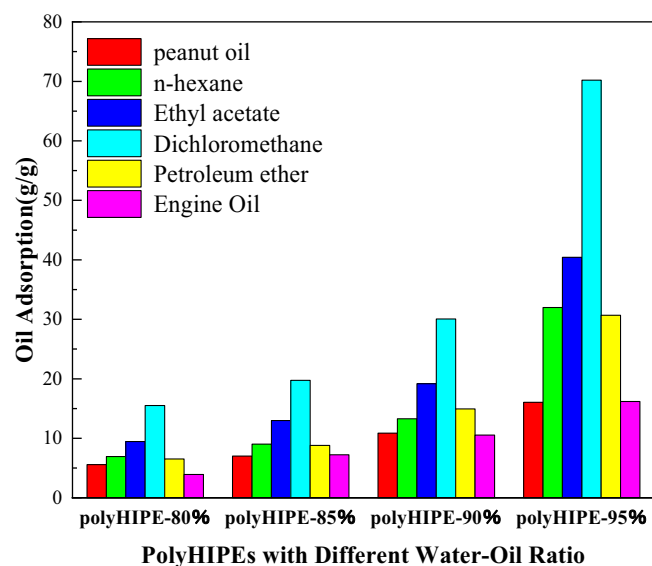


FIGURE 7 Saturated oil adsorption of polyHIPEs with different water-oil ratios: 80%, 85%, 90%, 95%. [Color figure can be viewed at wileyonlinelibrary.com]

(stained with methylene blue) and oil droplets (stained with Sudan II) on the polymer foam surface, respectively. It is observed that the water droplets remain spherical on the polymer foam surface, while on the contrary the oil droplets are rapidly adsorbed. In addition, the water contact angle of the polymer foam is 140° as shown in Figure 5c. For comparison, polystyrene foam with styrene: DVB = 3:1 (mole ratio) was prepared. From Figure S2, it can be found that the water contact angle of the polystyrene foam is 136° which is lower than the poly(*t*BS-co-DVB) foam. In order to study the actual oil-water separation performance of the polymer foam, the removal of oil spill and heavy oil under water experiments were performed by ethyl acetate and dichloromethane stained with Sudan II as oil models, respectively. The specific steps as shown in Figure 6 are as follows: The obtained polymer foam was first cut into small pieces which were put into a beaker containing ethyl acetate. It can be discovered that the polymer foam floated on the water surface due to its high porosity and low density. After the ethyl acetate on the surface of the solution was completely adsorbed by polymer foam, the polymer foam was clamped out with tweezers. As can be seen from Figure 6a, the solution in the beaker became clarified and the polymer foam pieces turned orange, indicating that ethyl acetate had been completely adsorbed by the polymer foam. Likewise, dichloromethane was first added to the beaker containing water. When the polymer foam was pressed down by tweezers, it can be noted that the polymer foam surface became like a silver mirror due to the hydrophobicity of the

polymer foam which results in the presence of an air layer between the polymeric surface and the water. When the polymer foam contacted with the oil at the bottom, the polymer foam immediately absorbed the oil within a few seconds while the water became clarified. The above experiments further prove that the prepared polymer foam has excellent lipophilicity and hydrophobicity, and has certain application prospects in sudden oil spill accidents and oily wastewater treatment.

3.4 | Oil adsorption capacity

The saturated oil adsorption performance of the polymer foam is an important parameter when dealing with oil-water separation problems. Hexane, ethyl acetate, methylene chloride, and petroleum ether are chosen as representatives of organic solvents, and peanut oil and motor oil as representatives of daily oil pollution. As shown in Figure 7 the oil adsorption performance of the polymer foam increases with the increase of the volume fraction, the adsorption capacity of the polymer foam reached to the peak with volume fraction of the internal phase of 95%. For hexane, methylene chloride, ethyl acetate, petroleum ether, peanut oil, and motor oil, the adsorption capacities were 32, 70.21, 40.43, 30.68, 16.06, and 16.2 g/g, respectively. This is because the polymer foam in this work is prepared using a water-in-oil HIPE template in which the functional monomer and crosslinker in the oil phase are polymerized to form the backbone and the water phase is removed to form the pores of the polymer foam. The greater the amount of water phase, the larger the formed pore structure, leading to larger oil adsorption capacity. However, it is worth noting that excessive internal phase volume fraction can form unstable emulsion and thus cause emulsion breakage and delamination. Therefore, we have only explored here an internal phase volume fraction of 95%. From the comparisons in Table 3, the poly(*t*BS-co-DVB) obtained in this work showed the higher oil adsorption capacity than the reported adsorbents.^{39–43}

3.5 | Oil retention and reusability

Whether the absorbing material can be recycled or not is very important in practical application, since it is about whether the absorbing material can comply with the needs of sustainable development. In addition, oil absorption material with a good recycling performance will have a longer using life, which will greatly save costs and bring higher treatment efficiency. In the adsorption-desorption process, hexane was used as the

TABLE 3 Comparison of oil/solvent adsorption capacities of various adsorbents.

Adsorbent	Components	Solvents	Adsorption capacity (g/g)	References
Resin (HIPE)	St/GMA/DVB/Fe ₃ O ₄ -PEI	Petroleum ether toluene	27.8 35.2	39
Sponge (HIPE)	St/DVB/Stearoylated cellulose	Dichloromethane	9.77	40
Sponge (HIPE)	St/DVB/Oyster shells	Dichloromethane ethyl acetate	43.2 30.9	41
Resin (HIPE)	VMA/LMA/DVB	Hexane, toluene	29.1 23.9	42
Resin (Suspension)	SMA/EHMA/DVB/St	Gasoline Diesel	10.1 10.7	43
Sponge (HIPE)	tBS/DVB	Hexane dichloromethane ethyl acetate motor oil	32 70.21 40.43 16.2	This work

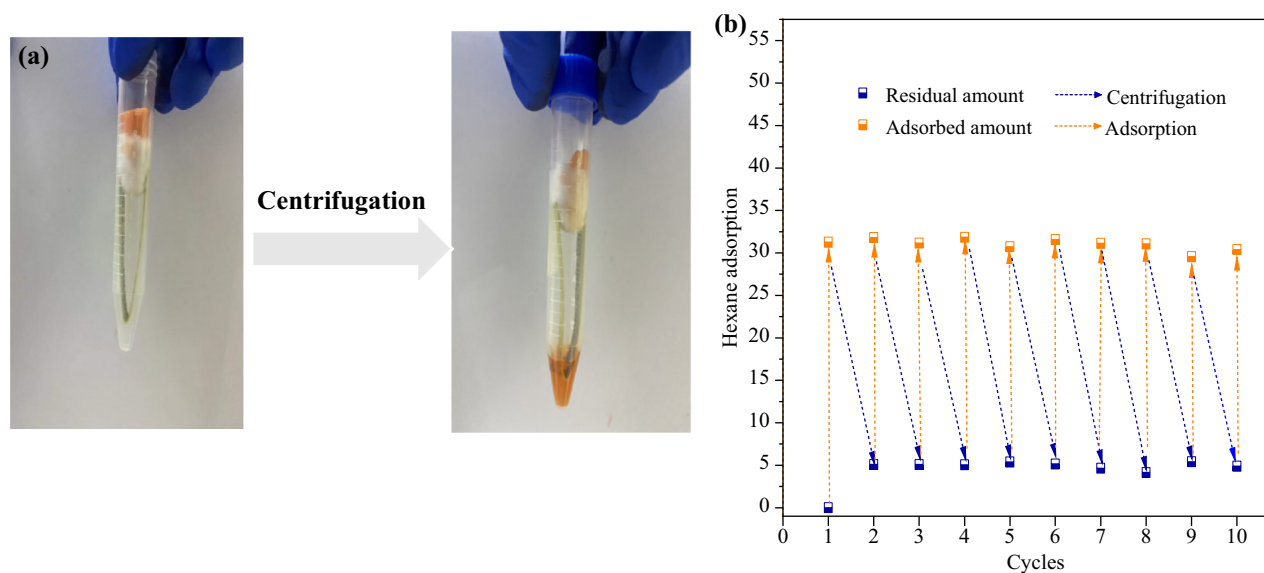


FIGURE 8 (a) Centrifugation for the extraction of n-hexane (dyed with Sudan II for clear observation), (b) oil adsorption capacities in different cycles. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

organic probe and polymer foam was centrifuged for desorption at 3000 rpm for 4 min. Based on the above equation, the oil retention of porous polymer was over 90% toward hexane can be calculated. The results show that the adsorbed organic solvent could locate inside the polymer foam without applying external force, while oil inside the porous polymer can be removed by simple centrifugation. Figure 8a illustrates the regeneration process of the polymer foam. The upper part of the centrifuge tube is the polymer foam cut into small pieces, the middle part is the cotton, and the lower part is the plastic support bent into a “U shape”. The plastic support is provided to prevent the centrifuged hexane from contacting with the polymer foam again, resulting in a weakened separation effect, while the middle part of the cotton is used as a buffer to prevent the lower part of the hard plastic support from colliding violently with the

upper part of the polymer foam during the high-speed centrifugation process, resulting in the destruction of the polymer foam. It can be observed that the color of the polymer foam is orange before centrifugation, and changed to light yellow after centrifugation and there is centrifuged hexane at the bottom of the tube. This indicates that most of the hexane inside the polymer foam is removed and only a small amount of hexane remains in the polymer foam. The regenerated polymer foam immediately undergoes the next adsorption, and the adsorption capacity for different number of adsorptions is shown in Figure 8b. After 10 adsorption-desorption experiments, the oil adsorption performance of the polymer foam only slightly decreased, which proves that the polymer foam has good recoverability and recyclability which makes it promising for practical oil adsorption applications.

4 | CONCLUSION

Porous poly(*t*BS-*co*-DVB) foams as prospective oil absorbing material for oil–water separation are successfully synthesized by HIPE template using 4-*tert*-butylstyrene (*t*BS) as hydrophobic monomer. The prepared polymer foam have interconnected porous structure, excellent hydrophobicity with a WCA of 140° and lipophilicity with oil contact angle of 0°. The optimization experiment results show that as the internal phase volume fraction increases from 80% to 95%, the pore size of the polymer foams gradually increase, meanwhile the SSA and porosity of the polymer foams increase from 33.06 to 102.89 m²/g and 84.07% to 92.94%, respectively. The oil adsorption capacity of poly(*t*BS-*co*-DVB) foams increase with the volume fraction of internal phase, as high as 70.21 g/g for dichloromethane, demonstrating excellent oil adsorption performance which is higher than related reports. This is because the interpenetrating pore structure inside the material further enhances the capillary effect, which is conducive to the transport and diffusion of oil inside the polymer foams and plays the role of storing oil molecules. The high porosity provides more internal space for oil storage, and therefore improve oil adsorption capacity. The oil adsorption performance of the porous polymer is well maintained even after several cycles of recycling. In conclusion, Poly(*t*BS-*co*-DVB) foam with the advantages of high porosity (92.94%), low density (0.0663 g/cm³), high SSA (102.89 m²/g), high absorption capacity (70.21 g/g toward dichloromethane) and excellent cycling performance have potential applications in the field of oil-water separation.

AUTHOR CONTRIBUTIONS

Yulai Zhao: Investigation (lead); resources (lead); supervision (lead); writing – review and editing (lead). **Lianzhi Shen:** Data curation (lead); formal analysis (lead); writing – original draft (lead). **Yu Yuan:** Writing – review and editing (supporting). **Longqiang Xiao:** Investigation (equal); resources (equal); supervision (equal). **Jingyu Cai:** Investigation (equal); supervision (equal). **Zhen Lu:** Investigation (supporting); supervision (supporting). **Linxi Hou:** Investigation (lead); resources (lead); supervision (lead).

ACKNOWLEDGMENTS

Thanks for the programs, including the Natural Science Foundation of Fujian Province (No. 2022J01569), National Natural Science Foundation of China (No. U21A200353), Major Program of Qingyuan Innovation Laboratory (No. 00122004), Key Program of Qingyuan Innovation Laboratory (No. 00221003) and ‘111’ Program of Fuzhou University, for financial supports.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Yulai Zhao  <https://orcid.org/0000-0002-2641-7192>

REFERENCES

- [1] S. Tadadjeu, H. Njangang, P. Ningaye, M. Nourou, *Afr. Dev. Rev.* **2022**, 34, 54.
- [2] V. Bove, C. Deiana, R. Nisticic^{1/2}, *J. Law Econ. Organ.* **2018**, 34, 272.
- [3] N. Antonakakis, J. Cunado, G. Filis, Gracia, *Resour. Policy* **2017**, 53, 147.
- [4] M. Rohal, C. Ainsworth, B. Lupher, P. A. Montagna, C. B. Paris, N. Perlin, P. M. Suprenand, D. Yoskowitz, *Environ. Model. Softw.* **2020**, 133, 104793.
- [5] R. Almeda, Z. Wambaugh, Z. Wang, C. Hyatt, Z. Liu, E. J. Buskey, *PLoS One* **2013**, 8, e67212.
- [6] C. H. Ainsworth, J. F. Samhouri, D. S. Busch, W. W. L. Cheung, J. Dunne, T. A. Okey, *ICES J. Mar. Sci.* **2011**, 68, 1217.
- [7] E. Zahedi, M. Vafaie-Sefti, M. Shadman, H. Naderi, M. Amiri, M. Noorbakhsh, *Pet. Sci. Technol.* **2017**, 35, 653.
- [8] T. Zhang, Z. Li, Y. Lü, Y. Liu, D. Yang, Q. Li, F. Qiu, *Chin. J. Chem. Eng.* **2019**, 27, 1282.
- [9] J. Yu, C. Cao, Y. Pan, *Adv. Mater. Interfaces* **2021**, 8, 2100061.
- [10] X. Dai, J. Lv, G. Yan, C. Chen, S. Guo, P. Fu, *Bioresour. Technol.* **2020**, 309, 123305.
- [11] U. Rojas-Alva, B. Skjonnning Andersen, G. Jomaas, *J. Environ. Manage.* **2019**, 250, 109470.
- [12] J. Pinto, A. Athanassiou, D. Fragouli, *J. Environ. Manage.* **2018**, 206, 872.
- [13] Z. Zhang, G. Sèbe, D. Rentsch, T. Zimmermann, P. Tingaut, *Chem. Mater.* **2014**, 26, 2659.
- [14] Z. Wang, X. Zhang, J. Gu, H. Yang, J. Nie, G. Ma, *Carbohydr. Polym.* **2014**, 103, 38.
- [15] J. Cheng, J. Hu, J. Wang, Z. Huang, B. Ye, S. Salvestrini, *Adsorpt. Sci. Technol.* **2022**, 2022, 1.
- [16] Y. Yang, H. Luo, H. Yang, H. Shi, J. Hou, *J. Polym. Sci.* **2021**, 59, 1456.
- [17] S. Wu, Z. Xing, Y. Yuan, W. Bai, L. Bao, L. Pei, H. Zhang, *Nanotechnology* **2021**, 32, 265706.
- [18] W. Wang, D. Yang, L. Mou, M. Wu, Y. Wang, F. Tan, F. Yang, *Colloids Surf., A Physicochem. Eng. Asp.* **2022**, 645, 128940.
- [19] P. Feng, X. Wang, J. Yang, *J. Mater. Sci.* **2022**, 57, 13097.
- [20] S. Fan, Z. Li, C. Fan, J. Chen, H. Huang, G. Chen, S. Liu, H. Zhou, R. Liu, Z. Feng, Y. Zhang, H. Hu, Z. Huang, Y. Qin, J. Liang, *J. Hazard. Mater.* **2022**, 433, 128808.
- [21] X. Li, Z. Yang, Y. Peng, F. Zhang, M. Lin, J. Zhang, Q. Lv, Z. Dong, *J. Hazard. Mater.* **2022**, 435, 129018.
- [22] S. L. Li, J. H. He, Z. Li, J. H. Lu, B. W. Liu, T. Fu, H. B. Zhao, Y. Z. Wang, *J. Hazard. Mater.* **2022**, 436, 129272.
- [23] Z. He, H. Wu, Z. Shi, X. Duan, S. Ma, J. Chen, Z. Kong, A. Chen, Y. Sun, X. Liu, *Colloids Surf., A Physicochem. Eng. Asp.* **2022**, 648, 129142.

- [24] H. Zhang, R. Zhao, M. Pan, J. Deng, Y. Biobased, Wu, *Ind. Eng. Chem. Res.* **2019**, 58, 5533.
- [25] C.-M. Yu, X.-H. Zhuang, S.-W. Zeng, Q.-X. Dong, Z.-X. Jing, P.-Z. Hong, Y. Li, *RSC Adv.* **2019**, 9, 17543.
- [26] Yu, C. Lin, W. Jiang, J. Jing, Z. P. Hong, Y. Li, *RSC Adv.* **2019**, 9, 37759.
- [27] Y. Zhu, Y. Zheng, F. Wang, A. Wang, *Chem. Eng. J.* **2016**, 284, 422.
- [28] Y. Yin, J. Pan, J. Cao, Y. Ma, G. Pan, R. Wu, X. Dai, M. Meng, Yan, *Chem. Eng. J.* **2016**, 286, 485.
- [29] O. Karatum, S. A. Steiner 3rd., J. S. Griffin, W. Shi, D. Plata, L. Flexible, *ACS Appl. Mater. Interfaces* **2016**, 8, 215.
- [30] S. Yun, H. Luo, Y. Gao, *J. Mater. Chem. A* **2014**, 2, 14542.
- [31] X. Xu, X. Li, Y. Chen, G. Liu, J. Gao, J. Liu, Z. Liu, S. Yue, L. Zhang, *J. Environ. Chem. Eng.* **2021**, 9, 106400.
- [32] J. He, H. Zhao, X. Li, D. Su, F. Zhang, H. Ji, R. Liu, *J. Hazard. Mater.* **2018**, 346, 199.
- [33] M. G. Perez-Garcia, M. C. Gutierrez, J. D. Mota-Morales, G. Luna-Barcenas, F. Del Monte, *ACS Appl. Mater. Interfaces* **2016**, 8, 16939.
- [34] N. R. Cameron, *Polymer* **2005**, 46, 1439.
- [35] G. Brusotti, E. Calleri, C. Milanese, L. Catenacci, G. Marrubini, M. Sorrenti, A. Girella, G. Massolini, G. Tripodo, *Polym. Chem.* **2016**, 7, 7436.
- [36] Y. Wang, U. Azhar, J. He, H. Chen, J. Zhao, A.-M. Pang, B. Geng, *Polym. Chem.* **2019**, 9, 40513.
- [37] A. Carranza, M. G. Perez-Garcia, K. Song, G. M. Jeha, Z. Diao, R. Jin, N. Bogdanchikova, A. F. Soltero, M. Terrones, Q. Wu, J. A. Pojman, J. D. Mota-Morales, *Mater. Interfaces* **2016**, 8, 31295.
- [38] K. X. Li, W. Chen, H. P. Yang, Y. Q. Chen, S. W. Xia, M. W. Xia, X. Tu, H. P. Chen, *Bioresour. Technol.* **2019**, 280, 260.
- [39] H. Huang, X. Li, C. Zhao, L. Zhu, Y. Li, Y. Wu, Z. Li, *J. Mater. Sci.* **2022**, 571, 336.
- [40] B. Pang, H. Liu, P. Liu, X. Peng, K. Zhang, *J. Colloid Interface Sci.* **2018**, 513, 629.
- [41] C.-M. Yu, X.-H. Zhuang, S.-W. Zeng, Q.-X. Dong, Z.-X. Jing, P.-Z. Hong, Y. Li, *RSC Adv.* **2019**, 931, 17543.
- [42] H. Zhang, R. Zhao, M. Pan, J. Deng, Y. Wu, *Ind. Eng. Chem. Res.* **2019**, 5814, 5533.
- [43] L. Kong, L. Ma, H. Jin, J. Hou, G. He, R. Zhang, *Polym. Adv. Technol.* **2019**, 306, 1441.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Y. Zhao, L. Shen, Y. Yuan, L. Xiao, J. Cai, Z. Lu, L. Hou, *J. Appl. Polym. Sci.* **2023**, e53801. <https://doi.org/10.1002/app.53801>